



JOBSHEET I

BASIC PROGRAMMING CONCEPTS

1. Purpose of the Practicum

After doing this practicum material, students will be able to:

1. Implement selection, looping, arrays, and functions in Java program code

2. Practicum

2.1 Selection

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions:

2.2.1 Election Practicum

Questions

1. Create a program to calculate the final score of students with the provision of 20% of assignment scores, 20% of quiz scores, 30% of UTS scores, and 30% of UAS scores. Each value entered has a value limit of 0 - 100. When a user enters outside of that range, an "invalid value" will come out. When the final score has been obtained, then convert the score with the following conditions:

Nilai Angka	Nilai Mutu		
	Nilai Huruf	Nilai Setara	Kualifikasi
$80 < N \leq 100$	A	4	Sangat Baik
$73 < N \leq 80$	B+	3,5	Lebih dari Baik
$65 < N \leq 73$	B	3	Baik
$60 < N \leq 65$	C+	2,5	Lebih dari Cukup
$50 < N \leq 60$	C	2	Cukup
$39 < N \leq 50$	D	1	Kurang
$N \leq 39$	E	0	Gagal

If the Letter Values obtained are A, B+, B, C+, C then PASS, if the letter values D and E then NOT PASS.

- Inputs from the program in the form of assignment value components, quizzes, UTS, UAS
- Output from the "invalid value" program if the value entered is outside the provisions
- The output of the program is in the form of final grades, letter grades, and the description of PASS/NOT PASS



Example of program running results

```

Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas: 85
Masukkan Nilai Kuis: 90
Masukkan Nilai UTS: 120
Masukkan Nilai UAS: 70
=====

nilai tidak valid
=====

```

```

Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas: 90
Masukkan Nilai Kuis: 40
Masukkan Nilai UTS: 75
Masukkan Nilai UAS: 85
=====

nilai akhir : 74.0
Nilai Huruf :B+
=====

SELAMAT ANDA LULUS
=====

```

Output :

The screenshot shows an IDE with the following code in the editor:

```

import java.util.Scanner;

public class ProgramMenghitungNilaiAkhir {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        double tugas, kuis, uts, uas;
        double nilaiAkhir;
        String huruf;

        System.out.println("Program Menghitung Nilai Akhir");

        System.out.print("Masukkan Nilai Tugas : ");
        tugas = input.nextDouble();

        System.out.print("Masukkan Nilai Kuis : ");
        kuis = input.nextDouble();

        System.out.print("Masukkan Nilai UTS : ");
        uts = input.nextDouble();

        System.out.print("Masukkan Nilai UAS : ");
        uas = input.nextDouble();

        // Menghitung nilai akhir
        if (tugas < 0 || tugas > 100 ||
            kuis < 0 || kuis > 100 ||
            uts < 0 || uts > 100 ||
            uas < 0 || uas > 100) {
            System.out.println("Nilai tidak valid");
        } else {
            nilaiAkhir = (tugas * 0.35 + kuis * 0.25 + uts * 0.25 + uas * 0.15);
            huruf = getHuruf(nilaiAkhir);

            System.out.println("Nilai Akhir : " + nilaiAkhir);
            System.out.println("Nilai Huruf : " + huruf);

            if (huruf.equals("A") || huruf.equals("B") || huruf.equals("C") || huruf.equals("D") || huruf.equals("E")) {
                System.out.println("SELAMAT ANDA LULUS");
            } else {
                System.out.println("Anda belum lulus");
            }
        }
    }

    public static String getHuruf(double nilai) {
        if (nilai >= 85) return "A";
        else if (nilai >= 75) return "B";
        else if (nilai >= 65) return "C";
        else if (nilai >= 55) return "D";
        else return "E";
    }
}

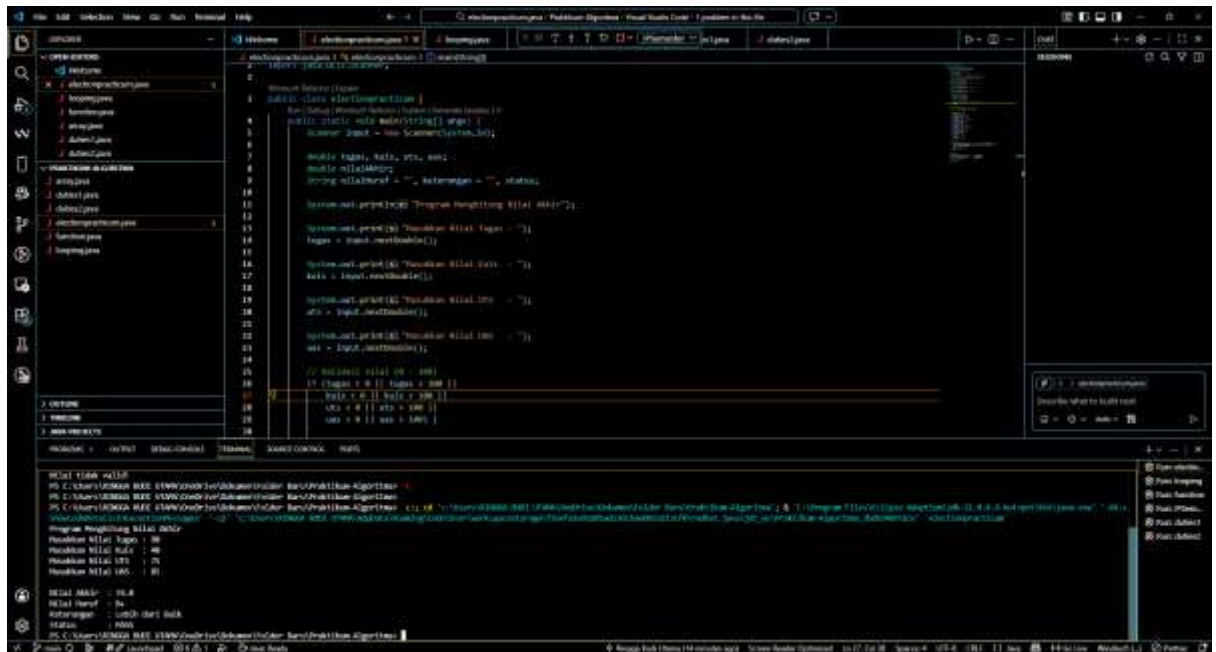
```

The output window shows the following results:

```

Program Menghitung Nilai Akhir
Masukkan Nilai Tugas : 85
Masukkan Nilai Kuis : 90
Masukkan Nilai UTS : 120
Masukkan Nilai UAS : 70
Nilai tidak valid

```



2.2 Looping

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions: **2.3.1 Repetition**

Practicum Questions

- Create a program that can display a row of numbers from 1 to n, with **n = the last 2-digit** of your NIM. Here are the conditions for printing the series
 - Number of multiples of 3 printed with symbols #
 - Even numbers other than multiples of 3 are printed according to the number
 - Odd numbers are printed with the symbol *
 - Numbers 10 and 15 are not printed

*if $n < 10$ then add 10 ($n += 10$)

Example 1: NIM input: 2541720102 then $n=12$

Output: ***2#4*#8#*#**

Example 2: NIM input: 2541720120 then $n=20$

Output: ***2#4*#8#*#*14*16*#20**

Output :

```

1 // Program: Hitung IP Semester
2 // Nama: [Nama]
3 // NIM: [NIM]
4
5 // Deklarasi Array
6 String[] namaMataKuliah;
7 double[] nilai;
8 int[] sks;
9
10 // Deklarasi Variabel
11 int jumlahMataKuliah;
12
13 // Input Jumlah Mata Kuliah
14 System.out.print("Masukkan jumlah mata kuliah: ");
15 jumlahMataKuliah = Integer.parseInt(scanner.nextLine());
16
17 // Inisialisasi Array
18 namaMataKuliah = new String[jumlahMataKuliah];
19 nilai = new double[jumlahMataKuliah];
20 sks = new int[jumlahMataKuliah];
21
22 // Loop Input Data
23 for (int i = 0; i < jumlahMataKuliah; i++) {
24     System.out.print("Masukkan nama mata kuliah ke-" + (i+1) + ": ");
25     namaMataKuliah[i] = scanner.nextLine();
26
27     System.out.print("Masukkan nilai ke-" + (i+1) + ": ");
28     nilai[i] = Double.parseDouble(scanner.nextLine());
29
30     System.out.print("Masukkan SKS ke-" + (i+1) + ": ");
31     sks[i] = Integer.parseInt(scanner.nextLine());
32 }
33
34 // Hitung IP Semester
35 double totalNilai = 0;
36 int totalSKS = 0;
37
38 for (int i = 0; i < jumlahMataKuliah; i++) {
39     totalNilai += nilai[i] * sks[i];
40     totalSKS += sks[i];
41 }
42
43 // Output IP Semester
44 double ipSemester = totalNilai / totalSKS;
45 System.out.println("IP Semester: " + ipSemester);
46
47 // Akhir Program
48

```

2.3 Array

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions:

2.4.1 Array Practicum

Questions

1. Create a program to calculate the Semester IP of the courses you took last semester. The formula for calculating the semester IP is as follows:

$$IP \text{ Semester} = \frac{\sum_i (Nilai \text{ Setara}_i * \text{bobot SKS}_i)}{\sum SKS}$$

Equivalent values are obtained from the following conversion table:

Nilai Angka	Nilai Mutu		
	Nilai Huruf	Nilai Setara	Kualifikasi
80 < N ≤ 100	A	4	Sangat Baik
73 < N ≤ 80	B+	3,5	Lebih dari Baik
65 < N ≤ 73	B	3	Baik
60 < N ≤ 65	C+	2,5	Lebih dari Cukup
50 < N ≤ 60	C	2	Cukup
39 < N ≤ 50	D	1	Kurang
N ≤ 39	E	0	Gagal



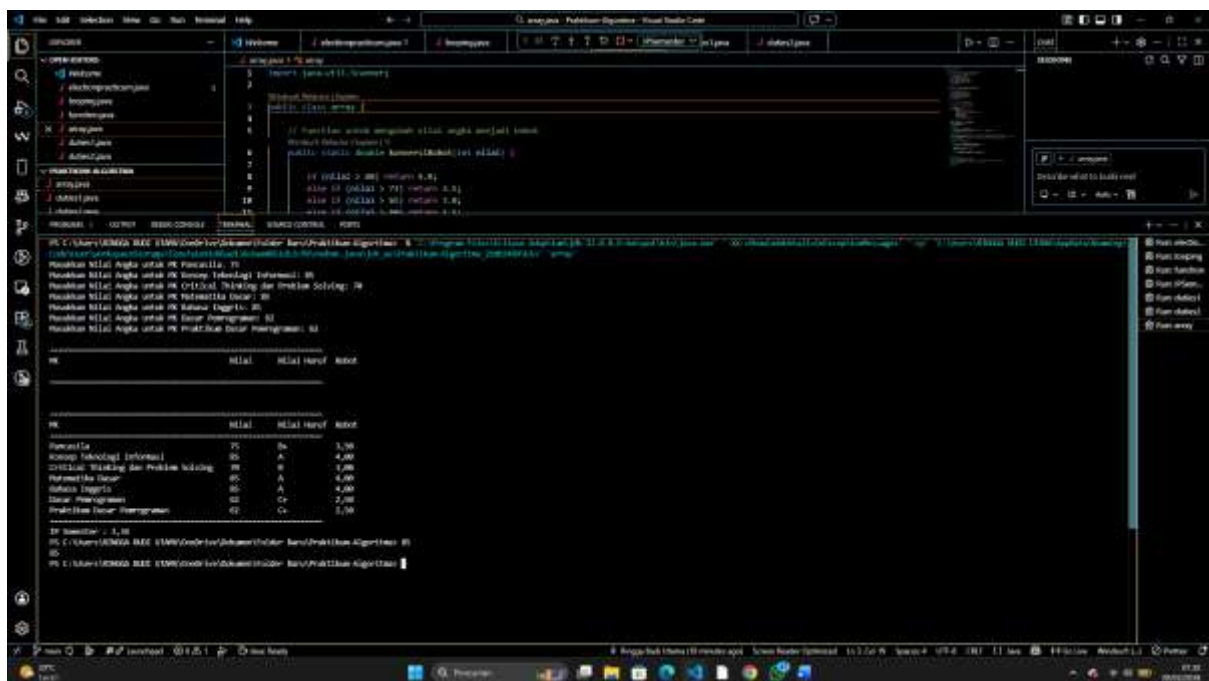
Input from the program is in the form of **course name**, **credit weight**, and **letter grades** of the course.

Example of program running results:

```

=====
Program Menghitung IP Semester
=====
masukkan nilai Angka untuk MK Pancasila: 75
masukkan nilai Angka untuk MK Konsep Teknologi Informasi: 85
masukkan nilai Angka untuk MK Critical Thinking dan Problem Solving: 70
masukkan nilai Angka untuk MK Matematika Dasar: 85
masukkan nilai Angka untuk MK Bahasa Inggris: 85
masukkan nilai Angka untuk MK Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Praktikum Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Keselamatan dan Kesehatan Kerja: 85
=====
hasil Konversi Nilai
=====
MK                                Nilai Angka    Nilai Huruf    Bobot Nilai
Pancasila                        75.00          B+             3.50
Konsep Teknologi Informasi       85.00          A              4.00
Critical Thinking dan Problem Solving 70.00          B              3.00
Matematika Dasar                 85.00          A              4.00
Bahasa Inggris                   85.00          A              4.00
Dasar Pemrograman                62.00          C+             2.50
Praktikum Dasar Pemrograman      62.00          C+             2.50
Keselamatan dan Kesehatan Kerja   85.00          A              4.00
=====
IP : 3.42
  
```

Output :





2.4 Functions

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course, so in this practicum, no experimental steps will be carried out. Answer the following questions:

2.5.1 Functional Practicum

Questions

RoyalGarden is a florist that has many branches. Every day the Flower Stock and the flowers sold are always recorded with the following details:

Row = Store Branch, Column = Flower stock on day x

	Aglonema	Yam	Alocasia	Rose
RoyalGarden 1	10	5	15	7
RoyalGarden 2	6	11	9	12
RoyalGarden 3	2	10	10	5
RoyalGarden 4	5	7	12	9

Details of the price of Aglonema = 75,000, Yam = 50,000, Alocasia = 60,000, Roses = 10,000.

- Create a function to display the income of each branch if all the flowers are sold out.
- Show the status of each branch with the following conditions:
 - If the branch's income > IDR 1,500,000, then the branch gets the status of "Very Good"
 - If the income ≤ IDR 1,500,000, the status is "Need Evaluation".

Output :

```

C:\Users\DELLA > cd C:\Users\DELLA\Documents\Praktikum\Praktikum Algoritma > java -cp . PraktikumAlgoritma
PraktikumAlgoritma
1. Total Pendapatan : Rp 1500000
Status : Very Good
2. Total Pendapatan : Rp 1500000
Status : Very Good
3. Total Pendapatan : Rp 1500000
Status : Need Evaluation
4. Total Pendapatan : Rp 1500000
Status : Very Good

```



3. Duties

Processing time: 100 minutes

1. Arrange the program to create two arrays and their contents as follows. The first array is a one-dimensional array of char CODE[10], containing the car's license plate code. The second array, the two-dimensional array char KOTA[10][12] contains the name of the city paired with the car's license plate code. The illustration of the display of the array is as follows:

A	B	A	N	T	E	N						
B	J	A	K	A	R	T	A					
D	B	A	N	D	U	N	G					
E	C	I	R	E	B	O	N					
F	B	O	G	O	R							
G	P	E	K	A	L	O	N	G	A	N		
H	S	E	M	A	R	A	N	G				
L	S	U	R	A	B	A	Y	A				
N	M	A	L	A	N	G						
T	T	E	G	A	L							

When the user provides a license plate code input, the program will remove the city name from the license plate code.

Output :

```

1  public class Main {
2      public static void main(String[] args) {
3          // Array nama kota
4          char[][] kota = {
5              {"A", "N", "T", "E", "N"},
6              {"J", "A", "K", "A", "R", "T", "A"},
7              {"B", "A", "N", "D", "U", "N", "G"},
8              {"C", "I", "R", "E", "B", "O", "N"},
9              {"B", "O", "G", "O", "R"},
10             {"P", "E", "K", "A", "L", "O", "N", "G", "A", "N"},
11             {"S", "E", "M", "A", "R", "A", "N", "G"},
12             {"S", "U", "R", "A", "B", "A", "Y", "A"},
13             {"M", "A", "L", "A", "N", "G"},
14             {"T", "E", "G", "A", "L"}
15         };
16
17         // Array kode
18         char[] kode = {"", "", "", "", "", "", "", "", "", ""};
19
20         // Input kode
21         Scanner scanner = new Scanner(System.in);
22         System.out.print("Masukkan kode plat: ");
23         String input = scanner.nextLine();
24
25         // Proses penghapusan
26         for (int i = 0; i < kode.length; i++) {
27             kode[i] = input.charAt(i);
28         }
29
30         // Output
31         System.out.println("Kode plat: " + String.valueOf(kode));
32     }
33 }

```



2. A program is used to store and manage a student's lecture schedule. Schedule data is stored in a string-type 2-dimensional array, provided that

- The line states the i-i schedule
- The column states schedule information: Course Name, Room, Lecture Day, Lecture Hours. Example :

```
schedule[0][0] = "Basic Programming"
schedule[0][1] = "Computing Lab 1"
schedule[0][2] = "Monday"
schedule[0][3] = "8:00-10:00"
```

- The number of lecture schedules is **n**, input by the user.

Create a function for

- Input lecture schedule data into a 2-dimensional array
- Display the entire lecture schedule in the form of a table
- Display lecture schedules based on specific days
- Display the lecture schedule by the name of a specific course

Output :

```

1 // Input lecture schedule data into a 2-dimensional array
2 // Display the entire lecture schedule in the form of a table
3 // Display lecture schedules based on specific days
4 // Display the lecture schedule by the name of a specific course
5
6 import java.util.Scanner;
7
8 public class Main {
9     // 2D array to store lecture schedule data
10    String[][] schedule;
11
12    // Constructor to initialize the array
13    public Main(int n) {
14        schedule = new String[n][4];
15    }
16
17    // Method to input lecture schedule data
18    public void input() {
19        Scanner scanner = new Scanner(System.in);
20        for (int i = 0; i < schedule.length; i++) {
21            for (int j = 0; j < schedule[i].length; j++) {
22                schedule[i][j] = scanner.next();
23            }
24        }
25    }
26
27    // Method to display the entire lecture schedule
28    public void display() {
29        System.out.println("Lecture Schedule");
30        for (int i = 0; i < schedule.length; i++) {
31            for (int j = 0; j < schedule[i].length; j++) {
32                System.out.print(schedule[i][j] + " ");
33            }
34            System.out.println();
35        }
36    }
37
38    // Method to display lecture schedules based on specific days
39    public void displayByDay(String day) {
40        for (int i = 0; i < schedule.length; i++) {
41            if (schedule[i][2].equals(day)) {
42                display();
43            }
44        }
45    }
46
47    // Method to display the lecture schedule by the name of a specific course
48    public void displayByCourse(String course) {
49        for (int i = 0; i < schedule.length; i++) {
50            if (schedule[i][0].equals(course)) {
51                display();
52            }
53        }
54    }
55
56    // Main method
57    public static void main(String[] args) {
58        Main main = new Main(10);
59        main.input();
60        main.display();
61        main.displayByDay("Monday");
62        main.displayByCourse("Basic Programming");
63    }
64
65 }

```

Output:

```

Lecture Schedule
Basic Programming Computing Lab 1 Monday 8:00-10:00
Basic Programming Computing Lab 2 Tuesday 10:00-12:00
Basic Programming Computing Lab 3 Wednesday 13:00-15:00
Basic Programming Computing Lab 4 Thursday 16:00-18:00
Basic Programming Computing Lab 5 Friday 19:00-21:00
Basic Programming Computing Lab 6 Saturday 08:00-10:00
Basic Programming Computing Lab 7 Sunday 11:00-13:00
Basic Programming Computing Lab 8 Monday 14:00-16:00
Basic Programming Computing Lab 9 Tuesday 17:00-19:00
Basic Programming Computing Lab 10 Wednesday 20:00-22:00

Search result to find out:
Basic Programming

```